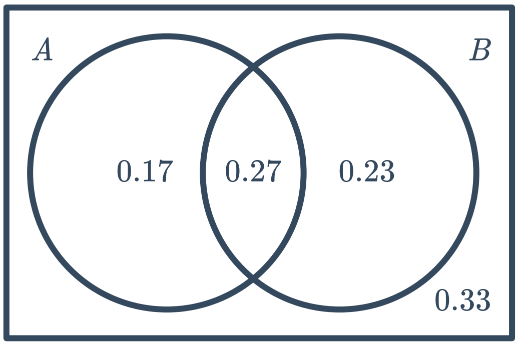
Conditional probability

1 Consider the following probability Venn Diagram. Find:

- a $P(A' \cap B)$
- b $P(A' \cup B')$



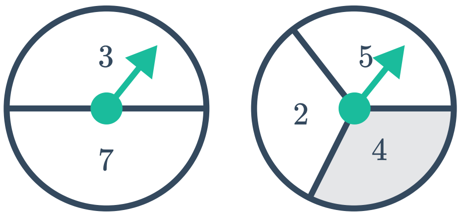
2 A and B are two random events with the following probabilities:
Find the value(s) of x if:

- $P(A) = 0.3 + x$
- $P(B) = 0.2 + x$
- $P(A \cap B) = x$

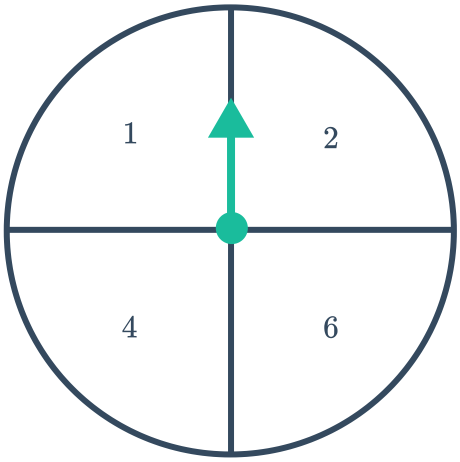
- a A and B are mutually exclusive.
- b A and B are independent.

3 The following two spinners are spun and the sum of their respective spins is recorded:

- a Construct a table to represent all possible outcomes.
- b Find probability that a 7 was spun given that the sum is less than 12.
- c Find the probability that the sum is odd given that a 4 was spun.



4 The following spinner is spun and a norma[redacted]sided die is rolled. The product of their respective results is recorded.



a Construct a table to represent all the possible outcomes.

b Find the probability that the product was a multiple of 4 given that a 2 was spun.

c Find the probability that a 6 was rolled given that the product was greater than 10.

d Find the probability that the product was greater than 4 given that the same number appeared on the dice and the spinner.

Dependent and independent events

5 Determine whether the following events of selection are independent or dependent.

a A card is randomly selected from a normal deck of cards, and then returned to the deck. The deck is shuffled and another card is selected.

b The selections of each ball in a lottery.

c Two cards are randomly selected from a normal deck of cards without replacement.

d Each student is allowed to randomly pick an item from the teacher's prize bag.

6 Two events A and B are such that $P\left(A\cap B\right)=0.02$ and $P\left(A\right)=0.2$.

a Are the events A and B are independent if $P\left(B\right)=0.1$?

b Are the events A and B are mutually exclusive?

7 Two events A and B are such that $P\left(A\cap B\right)=0.1$ and $P\left(A\right)=0.5$. Calculate $P\left(B\right)$ if events A and B are independent.

8 Consider $P\left(A\right)=0.2$ and $P\left(B\right)=0.5$:

a Find the maximum possible value of $P\left(A\cup B\right)$.

b Hence, are events A and B mutually exclusive?

c Find the minimum possible value of $P\left(A\cup B\right)$.

9 There are 4 green counters and 8 purple counters in a bag. Find the probability of choosing

10 In a game of Blackjack, a player is dealt a hand of two cards from the same standard deck. Find the probability that the hand dealt:

a Is a Blackjack. (An Ace paired with 10, Jack, Queen or King.)

b Has a value of 20. (Jack, Queen and King are all worth 10. An Ace is worth 1 or 11.)

11 In a game of Draw Poker, a player is dealt a hand of 5 cards from the same deck.

a Find the probability of being dealt a:

i Flush (five cards of the same suit)

ii Royal flush (10, Jack, Queen, King and Ace of the same suit)

b How many possible hands are there?

12 In a game of monopoly, two dice are rolled.

a Is the outcome of the second die dependent on the outcome of the first?

b If a 5 is rolled on one of the dice, find the probability of rolling a 5 on the second die.

13 In a lottery there are 45 balls.

a Find the probability of a particular ball being drawn first.

b A ball is discarded after it has been drawn. If ball number 35 is drawn on the first go, find the probability of ball number 20 being drawn next.

c Is the probability of each successive ball drawn the same as the probability of the first ball drawn?

d Are the draws dependent or independent events?

14 A standard deck of cards is shuffled and placed face down. Dylan attempts to draw five cards randomly so as to get a royal flush (ten, jack, queen, king, ace all of the same suit) .

a If he draws a queen of spades on the first go, find the probability that he draws a ten of spades on the second go.

b If he draws a queen of spades on the first go and a ten of spades on the second go, find the probability that he draws a jack of spades on the third go.

c Is the probability of each draw dependent or independent of the previous draw?

15 A number game uses a basket with 6 balls, all labelled with numbers from 1 to 6. 2 balls are drawn at random. Find the probability that the ball labelled 3 is picked once if the balls are drawn:

a With replacement.

b Without replacement.

16 A standard deck of cards is used and 3 car[illegible]e drawn out. Find the probability, in fraction form, of the following:

a All 3 cards are diamonds if the cards are drawn with replacement.

b All 3 cards are diamonds if the cards are drawn without replacement.

17 A number game uses a basket with 9 balls, all labelled with numbers from 1 to 9. 3 balls are drawn at random, without replacement. Find the probability that:

a The ball labelled 3 is picked.

b The ball labelled 3 is picked and the ball labelled 6 is also picked.

Tree diagrams and probability trees

18 A container holds four counters coloured red, blue, green and yellow. Construct a tree diagram representing all possible outcomes when two draws are done, and the first counter is:

a Replaced before the second draw.

b Not replaced before the next draw.

19 A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.

a Construct a probability tree of this situation with the correct probability on each branch.

b Find the probability of selecting two hearts.

20 Two marbles are randomly drawn without replacement from a bag containing 1 blue, 2 red and 3 yellow marbles.

a Construct a tree diagram to show the sample space.

b Find the probability of drawing the following:

i A blue marble and a yellow marble, in that order.

ii A red marble and a blue marble, in that order.

iii 2 red marbles.

iv No yellow marbles.

v 2 blue marbles.

vi A yellow marble and a red marble, in that order.

vii A yellow and a red marble, in any order.

21 Each year at Sicily High School, students t[redacted]art in a competition to recite the digits in π. Maria won in 2014 and 2015, but there's a new contender, Dario.

a If only these two compete, construct a tree diagram of the possible winners from 2014 to 2018 inclusive.

b If Maria and Dario are equally likely to win each year, find the probability that Dario wins in 2016 given that Maria won in 2018.

c Find the probability that Dario wins two of the three competitions from 2016 to 2018 given that Maria won in 2017.

22 When Fred gets ready for work in summer, he first decides whether it will be a hot day (H) or not (N) and then decides whether to wear a tie (T) or to just dress casually (C).
The chance of Fred deciding it will be a hot day is 0.7. If he decides it will be a hot day, there is a 0.2 chance he will wear a tie. If he decides it will not be a hot day, there is a 0.85 chance he will wear a tie.

a Construct a probability tree for this situation.

b Calculate the probability Fred decided it was a hot day given that he wears a tie. Round your answer to two decimal places.

23 In tennis if the first serve is a fault (out or in the net), the player takes a second serve. A player serves with the following probabilities:

- First serve in: 0.55
- Second serve in: 0.81

a Construct a probability tree showing the probability of the first two serves either being in or a fault.

b Find the probability that the player needs to make a second serve.

c Find the probability that the player makes a double fault (both serves are a fault).

24 Han draws two cards, without replacement, from a set of cards numbered 1 to 4 to create a 2-digit number. If he draws a 1 then a 2 the number 12 is formed.

a Construct a tree diagram to find all possible outcomes.

b Find the probability that the number formed is:

i Odd
ii Divisible by 11.
iii Even
iv Divisible by 3.
v Divisible by 5.
vi Divisible by 7.
vii Divisible by 4.

25 Nine pilots from StarJet and 7 pilots from [redacted]er offer to take part in a rescue operation. If 2 pilots are selected at random:

a Construct a probability tree showing all possible combinations of airlines from which the pilots are selected.

b Find the probability that the two pilots selected are from:

i The same airline

ii Different airlines

26 Sophia has three races to swim at her school swimming carnival. The probability that she wins a particular race is dependent on whether she won the previous races, as summarised below:

- The chance she wins the first race is 0.7.
- If she wins the first race the chance of winning the second is 0.8.
- If she loses the first race then the chance of winning the second is 0.4.
- If she wins the first two then the chance of winning the third race is 0.9.
- If she lost the first two then her chance of losing the third race is 0.9.
- If she won only one of the first two races, then the chance of winning the third is 0.6.

a Construct a probability tree to represent all outcomes in this situation.

b Calculate the probability Sophia won all three races correct to three decimal places.

c Calculate the probability Sophia won the third race, correct to three decimal places.